

More Than the Risk: Frontier LLM Behaviour in AI Development Races Depends on the Rival

Anonymous Author(s)
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Abstract

Large language models are increasingly used to advise and act on consequential decisions. As such systems become more involved in organisational decision making, an AI-safety question arises: how would frontier agents behave when competition rewards speed while caution reduces catastrophic risk? We study this question in an idealised AI-development race, where agents repeatedly choose between slower safe development and faster unsafe development.

Across the tested eligible endpoints, risk responses share a qualitative direction but differ in shape. Agents respond to increasing risk, with profiles ranging from gradual adaptation to near-deterministic switching. Behaviour also depends strongly on the opponent. In a scripted-rival study of five frontier-model endpoints that passed a frozen task-validity screen, changes in the rival's behaviour move the agents substantially, with a magnitude that varies by endpoint and can overlap the effect of changing the stated risk itself. Thus, self-play outcomes can reflect an interaction between two copies of a policy rather than an intrinsic risk preference.

Eligible endpoints also show a distinctive relationship to the theoretical and human benchmarks. An analysis extending the evolutionary comparison across the eligible panel places the four graded profiles closer in response shape to the weak-selection regime than to its strong-selection prediction, while their action trajectories generally occupy a more concentrated observed trajectory space than those of human participants: 13 of 15 eligible cells fall below every draw of a dyad-matched human null. These results suggest that AI-mediated competitive decisions may be model-specific, opponent-dependent and behaviourally concentrated, motivating evaluation at the level of interactions and trajectories rather than aggregate action rates alone. Claims are scoped to the tested endpoints, one prompt version and one idealised game.

Keywords

Large language models, AI development races, multi-agent systems, repeated games, AI safety, opponent-conditioned behaviour

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1 Introduction

LLMs are increasingly studied as agents for decision support and strategic interaction [2, 18]. As organisations consider delegating more strategic tasks to such systems, an AI-safety question follows: if a decision about how fast and how carefully to develop a powerful technology were delegated to a frontier model, what would the model decide, and would its decision look like the ones people make? The question is prospective throughout, and we do not claim that any organisation today hands a safety decision to a language model. We claim only that the behaviour a model would bring to such a decision is measurable now, in a controlled game. Controlled evaluation before widespread deployment can expose interaction effects that would be difficult to isolate in operational settings.

Competition over advanced artificial intelligence can create a safety dilemma: each company may benefit from moving faster, but safety work slows it down when rivals do not follow the same standard [6, 7, 11]. An *AI development race* is a simple game that represents this tension, in which every player chooses each round between slower Safe development and faster Unsafe development [12, 17]. The game is idealised and does not model a real AI company in full. Its value is control: it lets us hold the stated danger, the rival's behaviour and the relative progress fixed one at a time, which no observation of real firms allows. We reproduce the two-player game studied by Domingos and Han [14] and add a compatible multiplayer version based on Han et al. [17]. We call one hosted model endpoint at one fixed configuration an *endpoint*; the same model family can be served under different dated versions or provider settings.

Human experiments show why an average is the wrong thing to compare. In the two-player study of Domingos and Han [14], the preregistered comparison between the two higher private-risk conditions found no clear difference; exploratory analyses instead linked later Unsafe choices to the opponent's previous action, to whether the player was ahead or behind, and to the player's first action, so an early choice changes the later path of the interaction. A useful human and LLM comparison must therefore examine round-by-round trajectories over the first five rounds, rather than only the overall Unsafe rate.

Two problems make that comparison harder than it looks. The first is that a correctly formatted action is not an informed one. LLMs already play games of this kind as simulated participants, decision-making agents and game-playing systems [1, 2, 5], yet a model can return a valid action without tracking the state, can predict an opponent's pattern without acting on it [2], and can distort one simulated experiment while reproducing three others [1]; in a repeated game a single such action changes the next state and every later round. Neither a plausible trajectory nor a verbal strategy report is therefore evidence here that the agent tracked the game, and an aggregate UNSAFE rate close to the human one is

not evidence that it reproduces human decision dynamics or holds a general safety preference [8].

The second is that prompt-based evaluation is sensitive to presentation choices that do not change what the task asks. Formatting, option order, response-token identity, output-format requests, whitespace and persona all shift model output [25, 27–29, 33], and strategic choice is no exception: reversing action labels or reassigning their payoffs changes choice frequencies [18], decisions move with topic, actor and world framing [26], and performance falls when a familiar default mapping is replaced by a stated alternative [30]. We therefore test wording, answer order, arithmetic disclosure, narrative framing and opaque response codes separately, and do not treat one familiar presentation as a neutral measuring instrument.

Reproducible game-based frameworks attack the same difficulty from the other side. FAIRGAME standardises repeated-game experiments across models, languages and persona conditions against game-theoretic predictions [9]; later work adds payoff scaling, a Public Goods Game and supervised recognition of canonical strategies, and finds cooperation moving opposite to its evolutionary benchmark [19, 20]. What those designs hold constant is the opponent, because the opponent in them is another copy of the model under test.

One more result shapes what a delegate population should be compared against. Simulated populations recover broad contrasts while showing less heterogeneity than the people they stand in for, in markets, survey populations and persona-based simulation [8, 10, 13, 16], which is why position papers call social simulation defensible for exploratory use until validated against the population of interest [4]. Strategic-game evidence is similarly unsettled, with policies that shift with parameters and framing, heuristics applied more rigidly than people apply them, cooperation in place of the predicted equilibrium, and action sequences that resist reproduction from prompts alone [22, 23, 31, 32]. A set of delegates that agrees with itself more than people do is a different object from a delegate that is simply more or less cautious, so our human comparison reports within-population diversity and not only aggregate UNSAFE rates.

We therefore ask three questions about the delegate’s behaviour. *Risk*: how do frontier agents trade speed against safety as the stated chance of catastrophe rises? *Rivalry*: are those choices a property of the model, or do they emerge from the opponent the model happens to meet? *Humans and theory*: do the agents reproduce the strategic structure that evolutionary game theory predicts, or the one human participants display?

Each answer changes how the next should be read. The four graded endpoints respond gradually to the stated danger, while one endpoint replaces that trade-off with a switch from always fast to always careful. But where we put the agent against scripted code instead of a second copy of itself, the rival’s stance shifts play on every endpoint; its magnitude is endpoint-specific and can overlap the effect of changing the stated danger. Neither external reference shows that shape: at its usual strong-selection setting the evolutionary model predicts a switch where the agents deliver a gradient, and on raw first-five-round trajectories human participants use nearly the whole observed trajectory space while eligible endpoints generally use a small part of it.

Three contributions follow. First, a scripted-rival design for repeated-game evaluation and the critique it licenses. That an LLM’s self-play behaviour need not predict its behaviour against a different agent is known [21], and fixed opponents are standard practice in this literature [24]; what we add is the version that matters for a safety reading, that self-play can hide the response to the treatment the study is about, measured against an undisclosed rival whose policy is fixed by code. Second, a population-level comparison with human participants read on first-five-round trajectories rather than on aggregate rates, which is what makes the compression of observed trajectories visible at all. Third, an auditable race mechanism in which the engine and not generated text resolves every round, entered through a comprehension screen that scores answers to frozen questions against ground truth and never reads a race, so that the set of endpoints interpreted strategically was fixed by a rule that cannot see the behaviour being interpreted. An exploratory matched group-size comparison remains in the supplementary material because it covers one endpoint and cannot identify a pure group-size effect.

Every behavioural conclusion is scoped to its tested endpoint, prompt version, decoding contract, provider path and run period. The labels SAFE and UNSAFE denote actions inside the experimental game and are not measurements of general LLM safety, of a subjective risk preference, or of suitability for autonomous deployment, and nothing measured here is a harm, a real safety outcome, or an action taken by an organisation.

2 Preliminaries

We first specify the two-player game reproduced from Domingos and Han [14] (§2.1). We additionally evaluate an exploratory N -player extension; its mechanism and results are reported in the supplementary material. Figure 1 summarises the two-player instrument and the three empirical lenses used in the main results.

2.1 Two-player game

Two companies $i \in \{1, 2\}$ play a repeated race. In every round t , both simultaneously choose an action $a_i^t \in \{S, U\}$ (Safe, Unsafe) and commit to it before either learns the other’s choice, and only then does the round resolve (Figure 1a). Resolution has two parts. First, progress accumulates as $P_i^t = P_i^{t-1} + \sigma(a_i^t)$ with $\sigma(S) = 1.0$ and $\sigma(U) = 1.5$, so Unsafe always closes the race faster. Second, the round pays a stage payoff $\pi_i^t = \pi(a_i^t, a_{-i}^t)$, read off the fixed matrix (own action by row, opponent action by column, Figure 1b):

$$\pi = \begin{pmatrix} 1.0 & 0.6 \\ 2.4 & 2.0 \end{pmatrix}. \quad (1)$$

Unsafe strictly dominates Safe at the stage level (row 2 exceeds row 1 in both columns): the only cost of Unsafe development enters later, through accumulated private risk.

The race lasts a minimum of 5 rounds; after every completed round from round 5 onward it terminates with probability 0.2, giving an expected horizon $\mathbb{E}[T] = 9$. Neither company is ever told the horizon, so neither can condition on the current round being the last (Figure 1a). Termination resolves in two further steps. First, the company with the higher cumulative progress P_i^T wins a prize $B = 100$, split 50/50 on an exact tie. Second, and only for that winner or tied winners, an accumulated private setback risk is

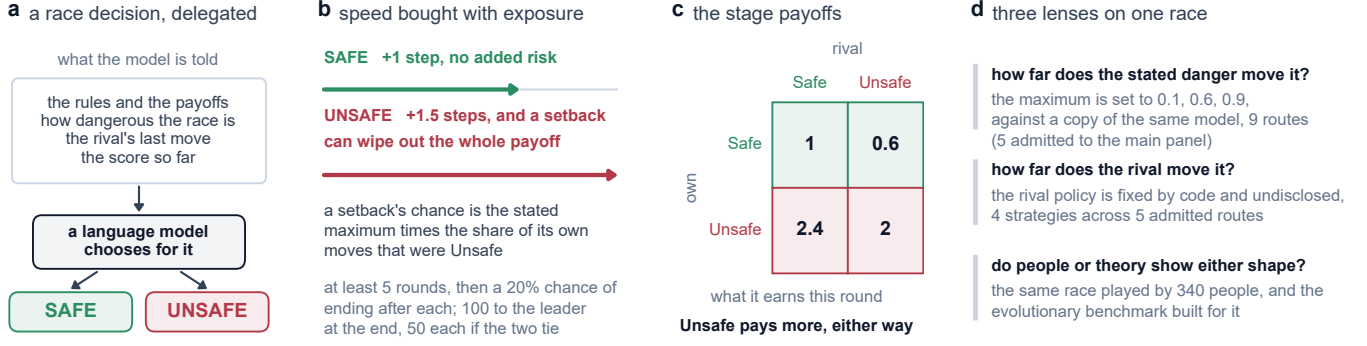


Figure 1: The game and the questions it lets us ask. (a) The model receives the rules, stated danger, rival’s last move and current score, then returns either **SAFE** or **UNSAFE**. **(b)** **SAFE** advances one step without added risk; **UNSAFE** advances faster but raises the chance that the winner’s payoff is wiped out. **(c)** The two-player stage-payoff matrix makes **UNSAFE** attractive against either rival move, so the delayed terminal risk supplies the safety tension. **(d)** The main results examine stated-risk response, scripted-rival response, and comparison with human trajectories and the evolutionary benchmark.

Table 1: The four benchmark strategies. Each is fixed by its opening move and by what it does afterwards, so AS and AU never react to the other player while CS and CAS react to nothing else. These four are also the fixed rivals of §4.2.

Strategy	Opening	Every later round
Always Safe (AS)	SAFE	SAFE
Conditionally Safe (CS)	SAFE	copy the rival’s last move
Conditionally Antisocial Safe (CAS)	UNSAFE	copy the rival’s last move
Always Unsafe (AU)	UNSAFE	UNSAFE

drawn:

$$q_i(T) = p_r^{\max} \cdot \frac{n_i^U(T)}{T}, q_i(T) \in [0, 1], p_r^{\max} \in \{0.10, 0.60, 0.90\}, \quad (2)$$

where $n_i^U(T)$ counts company i ’s Unsafe choices over the whole race and $p_r^{\max}$ is a treatment-level constant, assigned once per race, that we vary across three risk levels. If the draw against $q_i(T)$ fails, that company’s entire race payoff, accumulated stage payoffs plus the prize, is zeroed. A company that ends behind is never subject to the draw: it keeps its stage payoffs regardless of how much Unsafe it played, but receives no share of the prize (Figure 1a). This asymmetry, in which risk is built by both companies but realised only against whoever is ahead, is what makes Unsafe development a genuine gamble rather than a free speed advantage.

2.2 Game-theoretic benchmark

The two incentives oppose each other. Stage-level dominance rewards Unsafe immediately, while across the full race Unsafe choices raise the setback risk for a winner or tied winner, so the best action can depend on the risk condition, earlier actions, and the expected response of other players.

We use four simple strategies as a theory benchmark, defined in Table 1. Two of them ignore the other player and two of them do nothing but answer it, which is what makes the set a benchmark: it separates a fixed disposition from a reaction. In the reconstructed finite-population evolutionary model of Domingos and Han [14], the most common strategy changes from AU at $p_r^{\max} = 0.10$, to CAS

at 0.60, and to CS at 0.90: the theory predicts not one fixed amount of Unsafe play but a risk-dependent shift between unconditional speed, an aggressive opening and conditional restraint. We use this as a qualitative benchmark, not as a fitted model of LLM output, because prompted self-play trajectories are not samples from an evolutionary population.

3 Experimental design and evidence policy

3.1 Agent protocol and units of analysis

At each decision an agent receives a versioned prompt carrying the round, the assigned maximum private risk, the disclosed state, and the preceding revealed action profile when $t > 1$. Same-round decisions are sealed: every response comes from the same pre-action snapshot, before the engine resolves actions, progress, payoffs and risk, so the environment rather than generated text determines all transitions. Raw response, parsed action, retry and parser status, pre-turn state, terminal record, configuration and seed allocation are retained for every decision.

Sampling parameters here are provider-visible contracts, not intended settings. The run manifests record a requested temperature of 0.7 that the hosted SDK forwarded on no baseline endpoint, and no provider confirmed an effective value. Seed handling was endpoint-specific: the SDK stripped the requested seed on the Google endpoints and forwarded it, application unconfirmed, on the OpenAI and Anthropic ones.

Both seats of a race are the same endpoint, so every gameplay rate below is mirror-match play, until §4.2 fixes the rival instead.

The main behavioural outcome is whether an agent chooses Unsafe. The race, not the individual decision, is the main independent experimental unit, because decisions from one race depend on earlier states and are not separate random samples; uncertainty is grouped by race or matched repetition wherever possible. A cell’s rate pools its decisions, while a paired contrast is the mean over matched repetitions of the within-repetition difference in race-level rates, so each repetition counts once however long its race ran; the two therefore need not differ by exactly the printed contrast. A *parse failure* is a response that does not follow the required action

format, and because the fallback action changes later states, one marks the whole race as contaminated.

3.2 Evidence strata and admission

We separate confirmatory gameplay from diagnostic presentation checks. A *diagnostic pilot* is an exploratory test for possible effects or failures; it does not estimate a stable effect. A *confirmatory* study tests a question under a plan fixed before any result is inspected, with its protocol, endpoint, prompt, configuration, race count and exclusion rules recorded in the run manifest. We never pool the two strata.

One presentation check is confirmatory and is described in the supplementary material: on two endpoints, a fully crossed rerun retells the same game under a race story or as an abstract contest, and substitutes the opaque response codes *P* and *Q* for SAFE and UNSAFE under both code mappings. A task-validity screen gates which endpoints enter the behavioural comparison. It is an inclusion rule, not a behavioural outcome, and all gameplay results below are analysed without ranking endpoints by its scores.

3.3 Quality-control screen on the endpoints

An action that parses is not evidence that the agent followed the race. Before behavioural analysis, each endpoint therefore answered the same frozen bank of 20 probes over six task-validity domains, three times. An endpoint entered the study only if it cleared the fixed thresholds of 80% overall accuracy, 75% state reconstruction and 75% terminal scoring. Five of nine endpoints passed; the other four still played the same confirmatory baseline and remain available throughout the trajectory-diversity analyses for transparency and sensitivity checks.

The screen reads probes and never reads a race, so gameplay cannot change its verdict. Two eligible endpoints pass one screening domain at the minimum attainable score. We therefore use the screen only as a task-validity inclusion rule, never as a capability axis or a behavioural contrast. The supplementary material retains the admission table and all three gate scores; whether such a screen measures comprehension at all is the subject of a concurrent submission [3]; no result below rests on a screen score.

4 Results

These endpoints respond to the danger they are told about, and their behaviour also depends strongly on the rival. Four eligible endpoints have profiles closer in response shape to the weak-selection evolutionary benchmark; Claude Opus 5 is the step-like exception. Eligible endpoints generally occupy a more concentrated observed trajectory space than human participants do: 13 of 15 endpoint-risk cells fall below every draw of the primary dyad-matched human null. The four subsections take those results in turn. A screen-failing endpoint appears below only where it is named as excluded. Three- to five-player races, full model tables, predictive diagnostics and robustness checks are in the supplementary material.

4.1 How agents trade speed against safety

There is a shared directional response across the eligible endpoints. UNSAFE play is non-increasing with the stated maximum private risk on every eligible endpoint, while the shape varies: four of the

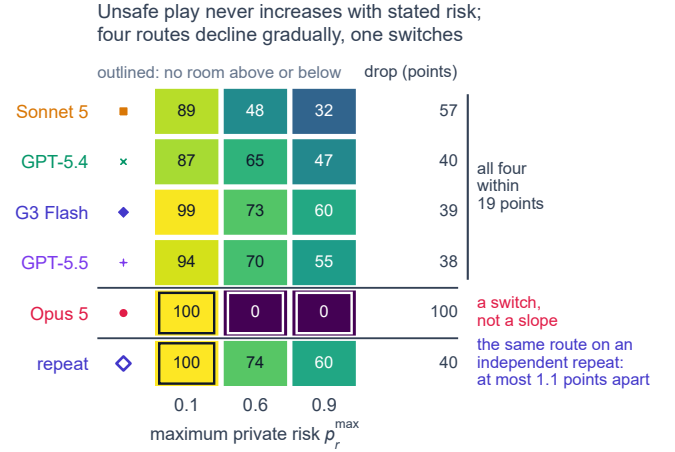


Figure 2: UNSAFE play is non-increasing with stated risk. The four graded endpoints fall within a 19-point band; Claude Opus 5 switches from the ceiling to the floor. The outlined cells are at a boundary and therefore leave no room for a larger or smaller observed rate. The repeat row is the same Gemini 3 Flash endpoint collected under an independent administration.

five give ground gradually and the fifth switches (Figure 2). Each endpoint supplied 30 races and 558 dependent within-race decisions under one frozen protocol, with zero parse failures and ten races per risk cell. Risk contrasts use matched repetition blocks, and race-weighted reanalysis preserves the low-to-high-risk contrast on every eligible endpoint (supplementary material).

The four graded endpoints, spanning three providers, have risk-response drops within a 19-point band, from 38.2 to 57.0 percentage points. We report that band rather than an average over all five, because averaging a switch with four slopes returns 54.8 points, a figure no graded endpoint is near and which one row produces.

That fifth endpoint follows a threshold-like policy. Claude Opus 5 plays UNSAFE on all 186 decisions at risk 0.1 and SAFE on all 372 decisions above it, with no race inside a cell departing from that. Its 100-point figure reaches the largest value on the scale, and because no risk level was run between 0.1 and 0.6, the location of its transition is not measured.

At the highest risk the game offers, where a winner’s whole payoff is zeroed with probability up to 0.9 in proportion to how much UNSAFE it played, the four graded endpoints still play UNSAFE on roughly a third to three fifths of their decisions against a copy of themselves, in the $r=0.9$ column of Figure 2. This variation shows that risk-aware behaviour remains contingent on endpoint and interaction context. Two endpoints can also reach a similar rate through very different sets of races, which \$4.4 measures directly.

For the repeated Gemini 3 Flash endpoint, the profile was stable across two administrations. Repeating the whole protocol on that endpoint under the same frozen contract moved no risk cell by more than 1.1 points over 186 decisions, against a 93.0-point spread in risk response across the nine endpoints, and moved its own risk response from 38.7 to 40.3. This bounds run-to-run variation for that endpoint at this sample size; it licenses no such bound for an

endpoint collected once, which matters here because no provider confirmed applying the requested sampling seed.

Representation-equivalent prompt changes also move rates. The directional risk response survives a fully crossed narrative-by-symbol rerun on two endpoints, although swapping the opaque action-code mapping shifts UNSAFE play by 8.3–9.5 percentage points (see the supplementary material).

4.2 The rival substantially reshapes play

In the five endpoints tested against scripted rivals, the observed behaviour does not support a context-independent risk profile. Across these endpoints, changing who is on the other side moves play substantially, but the magnitude varies by endpoint and can overlap the effect of changing how dangerous the race is told to be. Every rate reported so far is mirror-match play, in which the rival is the endpoint. One campaign fixes the rival instead.

The rival in that campaign is code, not a model. Five eligible endpoints each played the four benchmark strategies of §2.2 at all three risk levels: 60 cells, ten races each, 600 races and 5,580 endpoint decisions with no parse failure anywhere. It is never disclosed to the endpoint, and every one of its 5,580 recorded rival moves was recomputed afterwards from the endpoint’s own recorded history, which no cell failed. The endpoint’s seat is counterbalanced five and five inside each cell, and each contrast is taken inside a repetition, so the hidden horizon is differenced out. Because rival-policy assignment is fixed and exogenous within matched blocks, the direction of a difference between rival policies is identified. Conditional policies can still respond mechanically to the endpoint’s previous move, as specified by the fixed policy.

Coverage is complete across the five endpoints, but decoding contracts are not homogeneous across endpoints. Cross-endpoint differences in effect magnitude are therefore descriptive rather than controlled endpoint comparisons; the full contract provenance is reported in the supplement.

Table 2 puts the two candidate drivers of play on the same endpoints and the same matched blocks. Changing the rival while the stated danger is held fixed moves play by at least a quarter of an endpoint’s decisions in every one of the fifteen cells. Changing the stated danger while the rival is held safe moves play by about ten points on three endpoints and by half or more of their decisions on the other two. Neither effect dominates the other, because the largest risk contrast is larger than the smallest rival contrast and the two ranges overlap. What the table establishes is that the rival is never the small term, not that it is always the large one, and which of the two is larger is a property of the endpoint rather than of the game.

All five tested endpoints respond to the scripted opponent policy. UNSAFE play rises from the rival that always plays SAFE, through the two conditional rivals, to the rival that never plays SAFE, in fourteen of fifteen endpoint-by-risk cells, with strict ordering in twelve. The one reversal is GPT-5.5 at risk 0.1, where the CAS rival draws 100.0% and Always Unsafe is 97.8%, so this is a boundary case rather than a clean counterexample (Figure 3a). The paired difference between the two unconditional rivals is positive in every cell. All fifteen paired-block 95% intervals have positive lower bounds, with the smallest at +16.5 points (Figure 3b). The response to the stated risk

survives the change of design as well: against a fixed safe rival, no endpoint’s UNSAFE rate rises as the stated risk rises.

How strongly an endpoint answers its rival is itself endpoint-specific, and Table 2 separates two behaviours rather than two sizes of one behaviour. Gemini 3 Flash and GPT-5.4 answer the rival by about the same amount whatever the stated danger, so for them the rival and the danger are separate inputs. The other three answer it far more strongly once the danger is real than when it is negligible, so for them the rival is what the danger is read through. The level differs as well as the swing: against a rival that never plays UNSAFE, GPT-5.4 still does so about a third of the time at every stated danger, which no other endpoint does. These are separate endpoint samples, so intervals that fail to overlap are a conservative sign of a difference rather than a test of one.

A mirror-match rate is therefore an interaction outcome of one policy meeting itself, rather than a stand-alone measurement of how an endpoint treats risk. Self-play exceeds the fixed-safe rate in thirteen of fifteen cells; the two exceptions are Claude Opus 5 at the floor, lower at risk 0.6 (0.0% against 1.1%) and tied at 0.0% at risk 0.9. On Gemini 3 Flash at the lowest risk, the distance is a lower bound because nine of its ten mirror-match races have every decision UNSAFE. These results motivate treating mirror-match rates as interaction outcomes rather than endpoint-level risk preferences.

The bound on the finding is the design itself. Four reduced strategies are four points in a space of policies rather than a sample of rivals, so this establishes what an endpoint does against *these* rivals, on one game and one prompt version, and five commercial endpoints are not a sample from a population of models.

4.3 Weak selection better matches the graded response shape

The four graded endpoint profiles are closer to the source study’s human-fitted weak-selection regime than to its strong-selection reference setting (Figure 4b). The direct profiles remain the clearest shape comparison: Gemini 3 Flash gives 98.9%, 73.1% and 60.2%, and Claude Sonnet 5 gives 89.2%, 48.4% and 32.3% over 30 races each. At the reference setting of Domingos and Han [14], $\beta = 2$ and $\mu = \beta/Z$ lose 98 points inside a risk step of 0.005 (Figure 4a). The all-five fit shows that Claude Opus 5 is the qualitative exception: its profile is near-step rather than graded, even though its weak-selection RMSE is still lower than its reference RMSE.

In the evolutionary benchmark, the abrupt transition arises under strong selection and becomes graded under weak selection. We swept selection strength across $\beta \in [0.001, 10]$ crossed with three mutation rules, 90 stationary-distribution cells, each estimated from independent chains with the between-chain spread retained so that a badly mixed cell cannot be mistaken for a result. As β falls towards neutrality the predicted high-risk UNSAFE rate rises off the floor and the model’s own risk response becomes graded, which is the shape the endpoints show. At $\beta = 0.01$ with $\mu = 0.05$ the model predicts 87.3%, 63.9% and 38.0%, the closest cell in the entire sweep to both direct endpoints, within 9.6 percentage points of Claude Sonnet 5 and 15.4 of Gemini 3 Flash in root-mean-square error.

The weak-selection cell is not one we chose to fit: $\beta = 0.01$ and $\mu = 0.05$ are the values the source study reports as its own best fit to human play. The interpretation stays bounded. A stationary

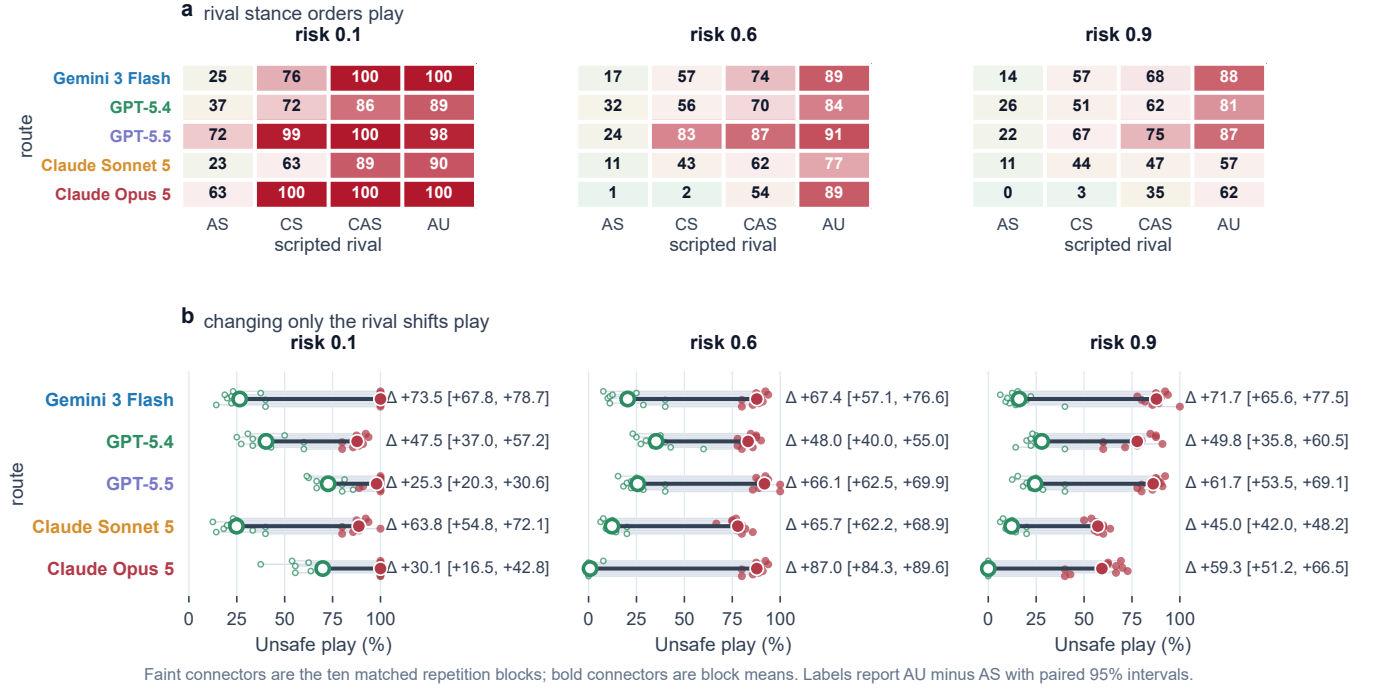


Figure 3: A shared opponent policy changes an endpoint’s play. Panel (a) is a response surface: each tile is an eligible endpoint’s Unsafe rate under one scripted opponent, with columns ordered from Always Safe to Always Unsafe and facets for the three maximum-risk conditions. Panel (b) shows the empirical response to changing only the opponent policy: open points are the Always Safe opponent and filled points are the Always Unsafe opponent, with faint matched repetition blocks behind each bold mean connector. Labels report the paired Always-Unsafe minus Always-Safe contrast with 95% intervals. Within-endpoint contrasts are the estimands; cross-endpoint magnitudes are descriptive under heterogeneous contracts.

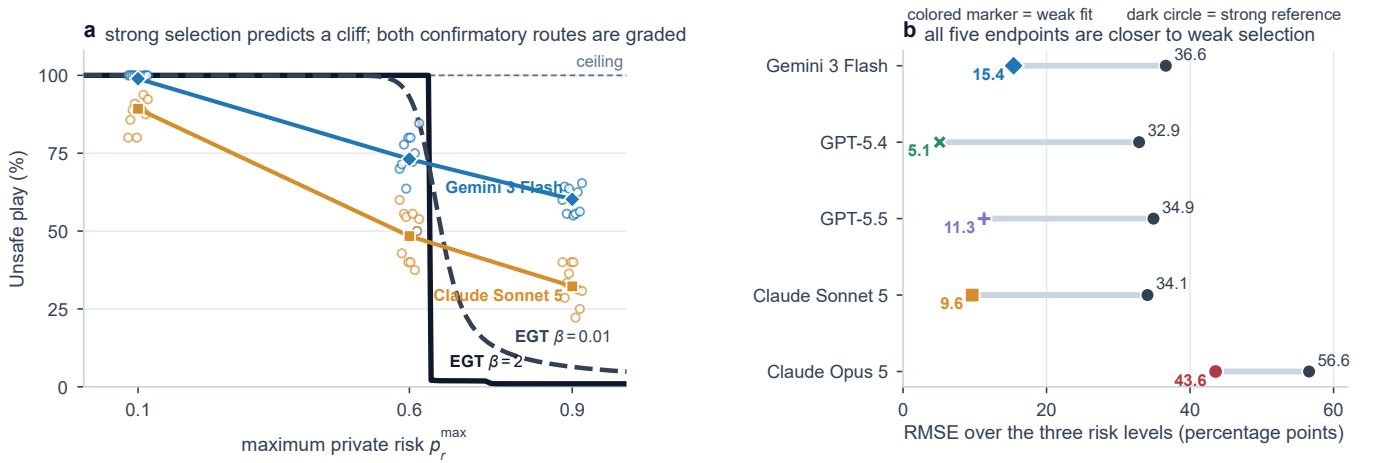
Table 2: What moves play: the rival’s stance against the stated danger, on the five eligible endpoints. The rival columns give the paired Always Unsafe minus Always Safe contrast at one stated danger. The danger columns give the paired risk 0.1 minus risk 0.9 contrast with the rival held at one policy. Every entry is in percentage points of UNSAFE play with a 95% interval over the same ten matched repetition blocks, so columns are comparable within an endpoint. Cross-endpoint magnitudes are descriptive, because decoding contracts differ across endpoints.

Endpoint	Rival stance, at a fixed stated danger			Stated danger, against a fixed rival	
	at 0.1	at 0.6	at 0.9	safe rival	unsafe rival
Gemini 3 Flash	73.5 [67.8, 78.7]	67.4 [57.1, 76.6]	71.7 [65.6, 77.5]	10.4 [5.4, 15.7]	12.2 [8.2, 16.1]
GPT-5.4	47.5 [37.0, 57.2]	48.0 [40.0, 55.0]	49.8 [35.8, 60.5]	12.3 [6.2, 18.6]	10.0 [5.9, 14.2]
GPT-5.5	25.3 [20.3, 30.6]	66.1 [62.5, 69.9]	61.7 [53.5, 69.1]	48.3 [43.1, 54.3]	11.9 [8.4, 15.2]
Claude Sonnet 5	63.8 [54.8, 72.1]	65.7 [62.2, 68.9]	45.0 [42.0, 48.2]	12.7 [8.5, 16.6]	31.5 [27.2, 35.3]
Claude Opus 5	30.1 [16.5, 42.8]	87.0 [84.3, 89.6]	59.3 [51.2, 66.5]	69.9 [57.1, 83.5]	40.7 [33.5, 48.7]

distribution describes strategy frequencies in a population under selection and mutation, while our measurements are prompted decisions in self-play; the agreement in shape says the graded risk response is consistent with weak selection and high exploration, not that the endpoints are evolving. What the sweep establishes is narrower and still useful: a claim that LLM agents contradict the evolutionary benchmark cannot rest on one setting of the selection strength, because the sign of the high-risk gap depends on it.

4.4 Eligible endpoints exhibit lower trajectory diversity than humans

Twenty people asked to run this race do not behave alike; twenty copies of a frontier endpoint very nearly do. Human participants arrive at their average level of UNSAFE play through many different histories, while each eligible endpoint reuses a small set of them, so an endpoint can match the human mean without resembling the human population at all. We measure this on first-five-round trajectories. Each player contributes one exact paired trajectory, a ten-bit string read within its own risk condition; because the



Panel a shows the two prespecified direct profiles; panel b uses the canonical secondary fit for all five eligible endpoints.

Figure 4: The evolutionary benchmark and the eligible endpoint profiles in a common risk coordinate. (a) The direct comparison keeps the two prespecified profiles, Gemini 3 Flash and Claude Sonnet 5, with individual race rates behind their means. The small-mutation limit is step-like at strong selection and graded near weak selection, while both direct profiles move gradually. (b) Each dumbbell compares the canonical RMSE over the three matched risk levels for the reference strong-selection regime ($\beta = 2$, $\mu = .02$) and the reported best weak-selection regime ($\beta = .01$, $\mu = .05$). All five eligible endpoints are closer to the weak-selection fit; Claude Opus 5 remains the qualitative near-step exception in panel (a).

progress increments are fixed, that string fully specifies the first-five-round joint-action trajectory. The comparison covers all nine endpoints that faced the screen, eligible and excluded alike, on the confirmatory baseline, 60 trajectories per endpoint, against 340 complete human trajectories from the de-identified dataset of Domingos and Han [14]. That single published study of a single game is the only human reference here, which bounds every human comparison below to that population and that mechanism.

Of 341 released participants, 340 provide complete first-five-round trajectories and 338 provide complete dynamic-model records. Among the complete trajectories, 168 source dyads contribute 336 trajectories; four complete trajectories have no complete partner after de-identification and are not used in the dyad null. Before using this dataset as an external reference, we validated the reconstruction by refitting the source study’s published dynamic specification on its released data: 2,888 participant-round observations from 338 participants in 172 race-pair clusters (see the supplementary material).

Figure 5 counts distinct first-five-round trajectories in matched samples of twenty, against 20,000 human resamples of ten complete dyads at each risk. The human sample is larger, so matching prevents sample size from inflating its diversity, while dyad sampling preserves the unit of interaction. Eighteen of 27 endpoint cells, including 13 of the 15 eligible cells, fall below every dyad draw. The participant-level sensitivity gives 21 of 27, including all 15 eligible cells; the primary dyad-null floors are 12, 15 and 13 distinct trajectories at risks 0.1, 0.6 and 0.9. The complementary participant-level table records effective counts of 18.9, 19.7 and 19.3 and mean pairwise distances of 0.47, 0.50 and 0.50. Panel b sorts the observed paths by frequency and panel c displays the modal five-round joint-action sequence. The same cells support two numerical checks reported

in the supplement: Hill $q=1$ counts effective trajectories and mean pairwise Hamming distance measures separation of their ten action bits. Under that complementary Hill $q=1$ analysis, every eligible endpoint sits entirely below the human interval at every risk level.

The concentration is strongest for Claude Opus 5, which produces a single trajectory, repeated by all 20 of its trajectories, at every risk level, while Claude Sonnet 5 uses two at risk 0.1. Concentration is also distinct from level: Gemini 3 Flash and GPT-5.5 finish within five points of each other at the highest risk and use a similar number of distinct trajectories, 12 and 13, yet the Gemini trajectories sit half again as far apart from one another, 0.41 against 0.27 in mean pairwise distance.

Under the participant-level sensitivity, two endpoints are the only exceptions, and the screen refuses both of them. GPT-5.4 mini uses 20, 17 and 17 distinct trajectories out of 20 at risks 0.1, 0.6 and 0.9, and GPT-5.4 nano 16, 19 and 16; both land inside the participant-level human null at every risk level, and no other endpoint reaches it once under that sensitivity. On this measure they are not less diverse than people. Under the primary dyad null, GPT-5.4 and GPT-5.5 also meet the finite-draw floor at risk 0.9, so the cluster-aware headline is 13 of 15 eligible cells rather than all 15. Because the two exceptions are outside the eligible set fixed by the inclusion rule of §3.3 before any race was read, they are reported, not interpreted as strategic diversity.

The same picture holds one level down, on the rates themselves. In the historical neutral-lane diagnostics, raw player-level UNSAFE rates plotted per risk condition in the supplementary material spread across nearly the whole available range for humans at every risk level while each plotted checkpoint occupies a narrower band of its own. Similarity between a checkpoint’s aggregate UNSAFE rate and the human mean therefore does not imply that it reproduces

the human distribution. A two-dimensional embedding of the same space is in the supplementary material; we rest no conclusion on it, because recolouring the same coordinates by outcome separates them almost entirely by UNSAFE rate.

Populations that look alike in rate can still be following different observed rules. Exploratory classifier and feature analyses in the supplement suggest that possibility, but use earlier pilot rosters containing neither GPT-5.4 nor GPT-5.5, so they support no claim about the eligible panel.

5 Discussion and conclusion

These agents incorporate both the stated risk and the rival’s stance. UNSAFE play is non-increasing as the stated chance of catastrophe rises on every eligible endpoint, while on the five endpoints where the rival was replaced by scripted code, the rival’s stance changes behaviour substantially, with a magnitude that varies by endpoint and can overlap the effect of changing risk. In this game, the five tested endpoints are therefore strongly context-sensitive: how the other side behaves can be more informative than the risk level alone.

Two things follow for how an agent of this kind should be measured. The first is that a rate collected in self-play carries the rival inside it, so it cannot be quoted as the endpoint’s own disposition, and a design that wants the disposition has to fix the rival by code and audit the moves that rival actually made. The second is that an aggregate rate is not a description of a population of agents. Twenty copies of an endpoint reach their average through far fewer distinct histories than twenty people do, so two systems can report the same headline rate while one of them has effectively one way of playing the game and the other has twenty. A safety argument that rests on a matched average is therefore resting on the weakest of the available comparisons.

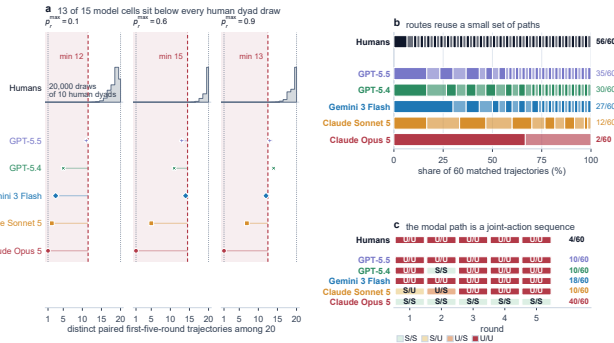


Figure 5: Human participants occupy a broader observed trajectory space than most eligible endpoint-risk cells under a dyad-clustered null. Panel a compares distinct paired first-five-round trajectories with 20,000 draws of ten complete human dyads; 13 of 15 eligible cells fall below every dyad draw. The participant-level null is a sensitivity analysis and places all 15 below every draw. Panel b sorts the 60 matched trajectories by path frequency; panel c shows the modal pooled joint-action path. Each tile is the focal endpoint’s action followed by its rival’s.

The human experiment is an external behavioural reference, not a target an agent should reproduce. Where an endpoint matches a human association, the two populations behave similarly on that one measured aspect of this task; it does not follow that they share a mechanism, and a mismatch does not invalidate the human result.

Four limitations bound the claims directly. Nine commercial endpoints are not a sample from a population of models, and any two of them differ in far more than the one property a comparison names, so every contrast drawn between endpoints in this paper is descriptive and none of it separates an endpoint’s behaviour from its provider, its scale or its training. The scripted-rival campaign covers five endpoints and four reduced strategies, which are four points in a space of policies rather than a sample of rivals. The human comparison rests on one published study of one game, 340 complete trajectories, and we do not add a second human dataset, because any other dataset would change the game as well as the population. And the evolutionary comparison is a reconstruction audited against pinned EGTtools source [15] rather than a bitwise reproduction of undisclosed author code, so it bounds the regime in which the benchmark is read rather than fitting the agents.

Four further limits are worth stating. The crossed presentation design covers two endpoints, enough to show the effect replicates but not to characterise endpoints in general. A matched group-size grid exists on one endpoint, but adding a company changes the stage payoff and the prompt length too, so no pure group-size effect is identifiable. Sampling was not under our control: no provider confirmed the requested temperature, and the seed was stripped on the Google endpoints and forwarded unconfirmed on the others. And this game models a private speed-versus-safety trade-off and nothing else, with no collective harm, regulation, communication or uncertainty about real capabilities.

Each of these limits points somewhere specific. The rival was fixed by code here because a fixed rival is what makes the contrast identifiable, and the natural next step is a rival that is itself an agent with a declared and audited policy, which would test whether the response survives against something that can adapt. The population result was read on one game, and reading it on a second game with the same trajectory measure would show whether the compression travels with the agent or with the task. And because the stated danger and the rival are separate inputs on two endpoints and not on the other three, an experiment that varies both together on more endpoints would say whether that split is a stable property of an endpoint or an artefact of the three danger levels this game offers.

Code, the de-identified run records and a checker that recomputes every reported quantity from them are provided with the supplementary material and will be released publicly with the final version.

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